AKADEMIKA

DCL-06

ASK / PSK / FSK

MODULATION / DEMODULATION KIT

EXPERIMENTAL MANUAL

…. A MESSAGE FROM

AKADEMIKA

Today's system designers are faced with tomorrow's problems. DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics.
It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, we are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

Sr. No. 15/8/1, Unit No.9, Kruti Industrial Estate, Opp.
Sangam Press, Kothrud, Pune, Maharashtra- 411038
INDIA
Contact No.:+91-9004904462
+91-7447438443
Email: lab@akademika.in
www.akademika.in

SAFETY RULES

Carefully follow the instructions contained in this handbook as they give important points on safety during the installation, use and maintenance. Keep this handbook always with you for any further help.

After the unpacking, set all accessories in order so that they are not lost and check the kit integrity. In particular, check that the kit is integral and shows no visible damage.

Before connecting the power supply to the kit, be sure that the jumpers and patch chords are connected correctly as per experiment.

In DCL-06 kit, signals are to be monitored with an oscilloscope or with help of indicators as explained in the manual.

This kit must be employed only for the use for which it has been conceived, i.e. as educational equipment and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault/bad operation, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity**.**

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kits has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

INDEX

TECHNICAL SPECIFICATIONS

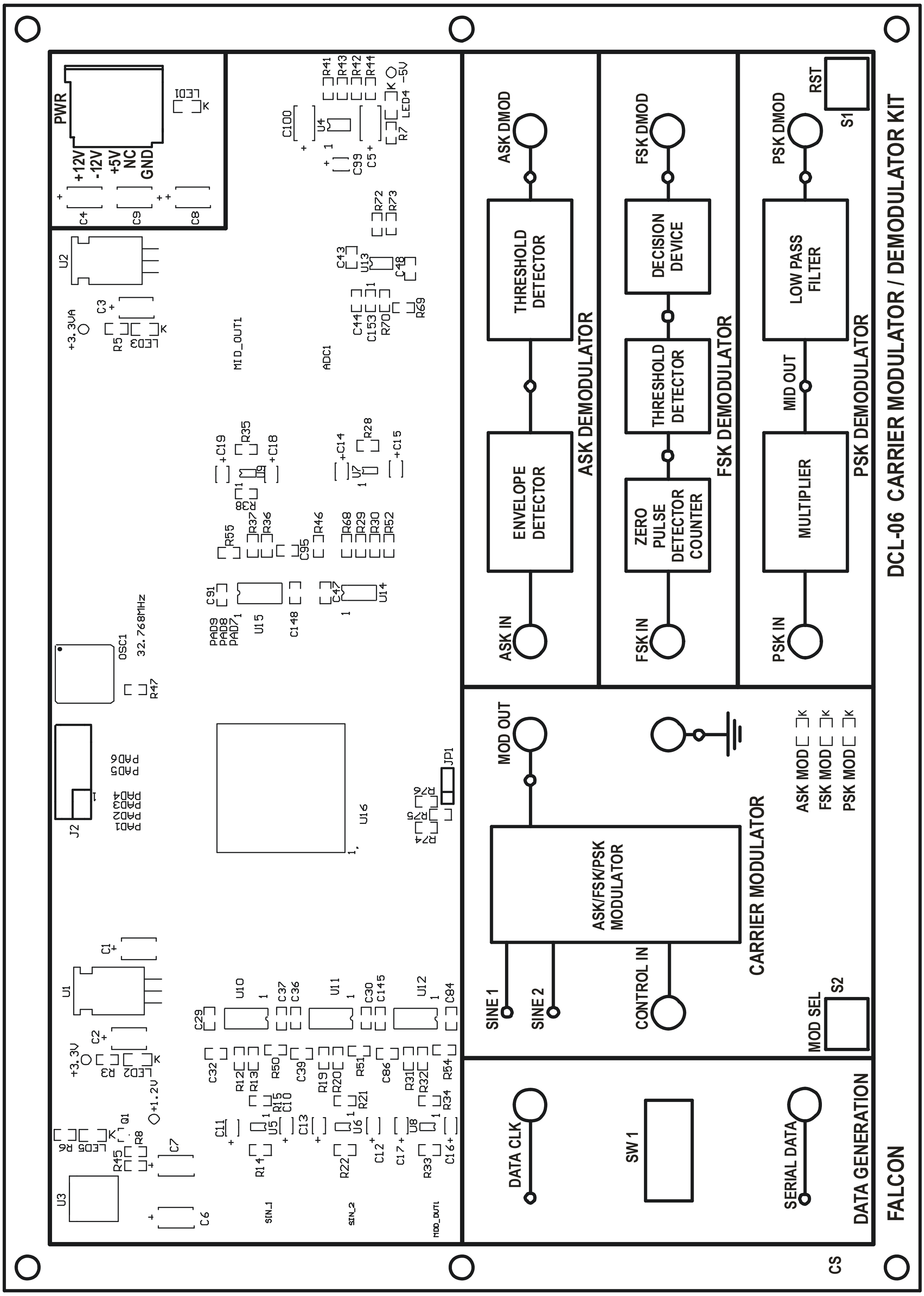
DCL-06: CARRIER MODULATOR / DEMODULATOR KIT.

DATA SIMULATOR	:	Onboard 8-bit variable NRZ-L pattern.
CRYSTAL OSCILLATOR	:	32.768 MHz
DATA CLOCK	:	256 KHz.
ON BOARD CRRIER SINE WAVES	:	1.024 MHz (0°), 1.024 MHz (180°), 512 KHz (0°).
CARRIER MODULATION	:	ASK, FSK, PSK.
CARRIER DEMODULATION	:	ASK, FSK, PSK.
INTERMEDIATE SIGNAL	:	Provision for observing intermediate signals during demodulation.
INTER CONNECTION	:	2 mm banana socket.
POWER SUPPLY	:	+12V, -12V, +5V, GND.
TEST POINTS	:	17.

ACCESSORIES

	DCL -06	Quantity
1	Short Links PCS2 Patch cord Male To Male	4
2	DCL-06 EXPT Manual	1
3	Power Supply (DCS)	1
4	USB Cable	2
5	Power cord	1

FUNCTIONAL BLOCK



INTRODUCTION

“DCL-06 initiates the user to the various Carrier Modulation Techniques, normally adopted in practice

“DCL-06”: CARRIER MODULATOR / DEMODULATOR KIT.

FEATURES:

1) The trainer comes with an onboard data simulator, which generates the NRZ-L pattern depending on positions of 8-bit switch and the reference clock (256 KHz), which enables the trainers to work in a stand-alone mode.

2) Three carrier modulation options are available on the kit.
   a) Amplitude Shift Keying (ASK)
   b) Frequency Shift Keying (FSK)
   c) Phase Shift Keying (PSK).

3) Onboard carrier generation circuit generates sine waves synchronized to the transmitted data. The three carrier sine waves generated onboard are of frequencies.
   a) 1.024KHz (0°)
   b) 1.024MHz (180°)
   c) 512KHz (0°)

4) Following options are available for detecting the information from the carrier:
   1) Envelope Detection for ASK.
   2) FSK Detection using Zero pulse Detector, Zero pulse Counter and Threshold Detection techniques.
   3) Squaring Loop Detection for PSK.

5) Interconnection facilities:
   Sockets and patch chords are provided for on board connections.

7) Test Points:
   All relevant test points are brought-out for observations. Observations are carried out on an oscilloscope.

8) Power Supply:
   -12V, +12V, +5V, GND.

PRINCIPLES OF DIGITAL MODULATION AND DEMODULATION TECHNIQUES:

Digital Communication is the technique by which the whole information is represented in terms of binary digits, i.e., `ones' and `zeros'. These digits are represented by discrete voltage levels and the clock frequency of a digital communication scheme is generally low. For long distance transmission, the data is made to modulate a continuous wave (sine wave) carrier. These techniques are called Carrier Modulation Techniques. The various types of Carrier Modulation Techniques normally adopted in practice fall under three broad categories.

a) Amplitude Shift Keying (ASK),
b) Frequency Shift Keying (FSK),
c) Phase Shift Keying (PSK).

a) AMPLITUDE SHIFT KEYING (ASK):

For all types of Carrier Modulation, the Carrier frequency should be at least 2 times base band of the modulating signal.

Assuming that the modulated carrier is a sine wave represented by the following equation.

C (t) = A (t) cos wt
Where C (t) = Carrier sine wave.
A (t) = Time varying amplitude
wt = Time varying angle.

Now in amplitude shift keying, the carrier is being transmitted only when the modulating data is `one' and when the data is `zero' the carrier is rejected from transmission. Thus the resulting modulated output of this type of modulation can be represented as follows:

M (t) = r (t) * C (t)
Where M (t) = Modulated Carrier
r (t) = Time varying modulating data which is either `one' or `zero'
Now M (t) = A (t) cos wt when modulating data is one.
0 when modulating data is zero.

An Envelope detector is used to recover the data from the modulated carrier.

b. FREQUENCY SHIFT KEYING (FSK):

In this type of modulation, the modulated output shifts between two frequencies for all `one' to `zero' transitions.

Let the carrier frequencies be represented by w1 and w2, and then we have:

M (t) = A (t) Cos w1t if data is `One'
A (t) Cos w2t if data is `Zero'.
Where A (t) = Time varying amplitude of the sinewave.
M (t) = Modulated carrier.

FSK Demodulator employs PLL logic for the recovery of data.

c. PHASE SHIFT KEYING (PSK):

In the PSK modulation or phase shift keying, for all `one' to `zero' transitions of the modulating data, the modulated output switches between the in phase and out of phase components of the modulating frequency. If the modulated carrier is represented by:

M (t)	=	A (t) Cos (wt + Phase)
Where A (t)	=	Time varying amplitude, and
wt	=	Time varying angle.
M (t)	=	Modulated carrier.

Then the phase equals 0 degree whenever the data equals `one' and the phase equals 180 degrees whenever the data equals 'zero'. This type of phase shift keying is called Binary phase Shift Keying (BPSK).

There are also other forms of phase shift like Quadrature Phase Shift Keying (QPSK), Differential Phase Shift Keying (DPSK) etc.

In QPSK, phase has 4 different values like 0 degree, 90 degrees, 180 degrees and 270 degree. They are used in the carrier modulation of differentially coded dibit pair etc. For the modulation of serial PCM data, BPSK is sufficient.

The BPSK detector works on the principle of square law. BPSK modulated carrier can be represented mathematically as:

M (t)	=	A (t) Cos (wt + Phase)
We have phase	=	0 degree - when data is 1
	=	180 degrees - when data is 0
Thus,		
M (t)	=	A (t) Cos wt - when data is 1
	=	-A (t) Cos wt - when data is 0

On squaring the modulated carrier, the negative sign gets eliminated and the frequency gets multiplied by two.

M^2 (t)	=	A^2 (t) cos2wt - for both data is 1 and 0
M^2(t)	=	A^2 (t) (1+cos2wt)/2

Now the in phase reference carrier can be recovered by dividing the frequency of the squared modulated carrier by two.

Once the carrier is recovered, the data can be detected by comparing the phase of the received modulated carrier with the phase of the reference carrier.

Circuit Description

- **Clock And Data Generator**

This block generates NRZ-L data of a variable pattern using VLSI technology depending on positions of eight bit switch SW1 also Reference clock of frequency 256KHz (DATA CLK) is generated. The crystal oscillator generates a 32.768MHz clock, from which a 256 KHz clock is recovered by using 10-bit binary down counter. For data generation output of SW1 and Load/Shift pulse of 32KHz is given to Parallel to serial shift register. Output of shift register is taken as Reference Data.

- **Carrier Generator**

Refer sheet 2 of DCL-06 for the Circuit diagram of carrier generator. The Carrier Generator consists of two sections that are SINE1_DAC and SINE2_DAC. SINE1_DAC generates 1.024MHz (0º) carrier and SINE2_DAC generates 512 KHz and 1.024 MHz (180º) carrier according to the selection of modulation scheme by switch S2. Digital input is fed by the FPGA to these DACs as per internal logic written in VHDL. The operating mode of IC THS 5641A is mode 1 which takes input in the form of twos complement. For mode 1 pin no. 25 of each IC U10 and U11 is connected to the DVDD through a resistor. Clock of 32.768MHz is fed by the FPGA to pin no. 28 of U10 and U11. The THS5641A (U10 and U11) delivers complementary output currents IOUT1 and IOUT2. IC OPA 690 (U5 and U6) are used as current to voltage converters. IC U5 and U6 give the SINE 1 and SINE 2 carrier at pin no.1 respectively.

- **Carrier Modulation**

Refer sheet 5 of DCL-06 for the Circuit diagram of carrier modulation. The Carrier Modulator consist MOD_OUT_DAC section. This section gives modulated signal at its output according to the selection of modulating techniques by switch S2. Carrier modulation is done by 2:1 digital multiplexer. The two inputs of digital multiplexer are SINE 1 and SINE 2 and select input is CONTROL IN. The logic for carrier modulation is written in the VHDL according to this logic digital input is fed by the FPGA (U16) to the DAC THS 5641A (U12). IC U12 also works in mode 1 configuration. The final modulated output is given by the IC OPA 690 (U8) at pin no.1.

- **ADC section**

Refer sheet 3 of DCL-06 for the Circuit diagram of ADC section. This section consists of IC TL082 (U13) as Low Pass Filter to remove high frequency components from the modulated signal, IC OPA 690 (U7) as DC shifter to give 2V dc shift to the filter output and IC ADS 830 (U14) for A to D Conversion. This digital output of IC U14 is given to the FPGA (U16) for demodulation.

- **ASK Demodulator section**

Refer sheet 5 of DCL-06 for the Circuit diagram of ASK demodulator. MID_OUT_DAC section consist of 14 bit DAC THS 5671A (U15) and OPA 690 (U9). The ASK modulated input is fed to the full wave rectifier. ASK demodulation employs ENVELOPE DETECTOR logic which is written in VHDL. According to the selection of modulating techniques MID_OUT_DAC section is used for ASK demodulator and PSK demodulator. MID_OUT_DAC section gives rectified output for ASK demodulation. The THRESHOLD DETECTOR detects the threshold level, which is used to recover the original amplitude levels of the data. So whenever the sine wave is transmitted, the detector identifies it as a `one' and whenever the carrier is absent, the detector identifies it as a `zero'.

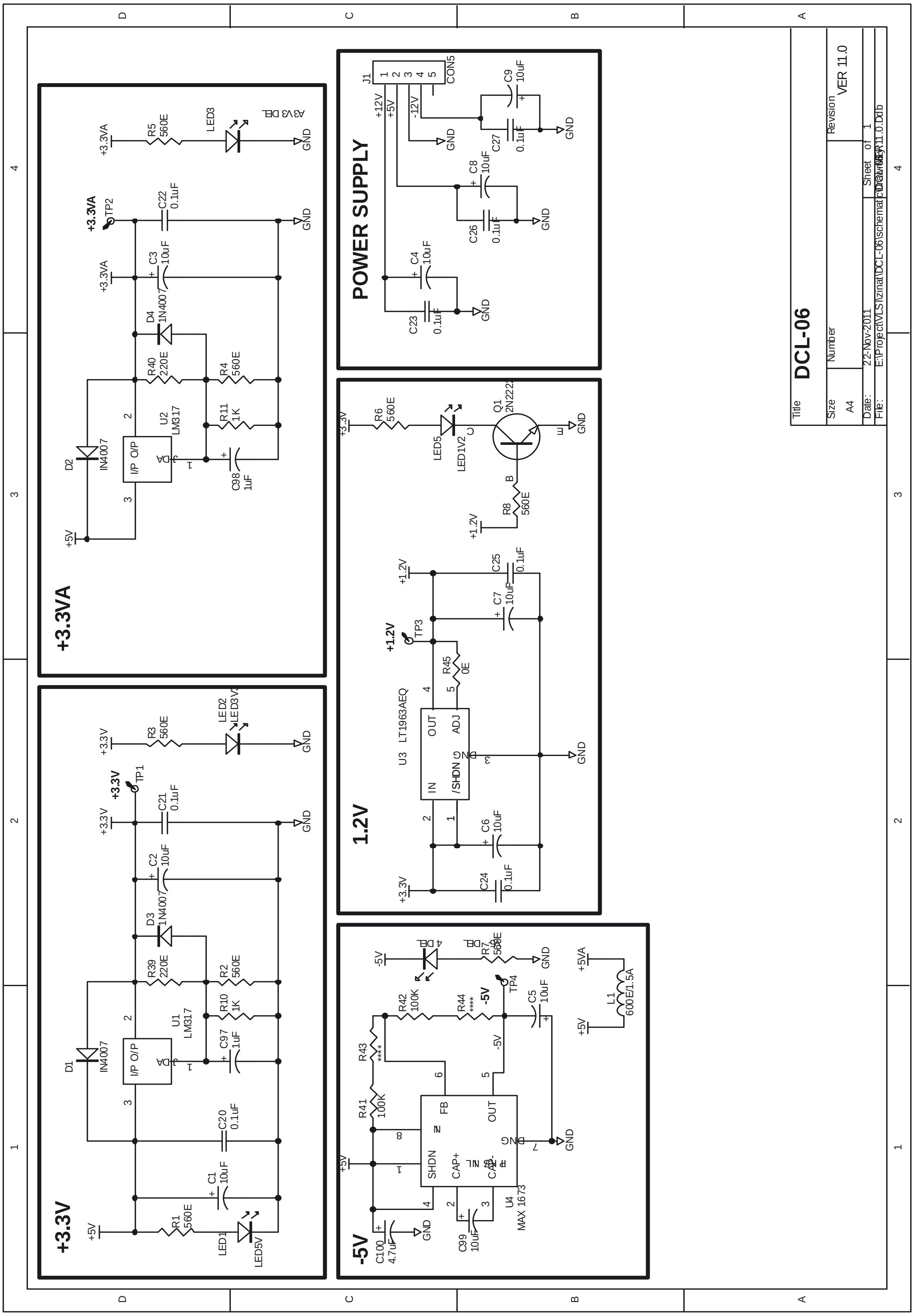
- **FSK Demodulation Section**

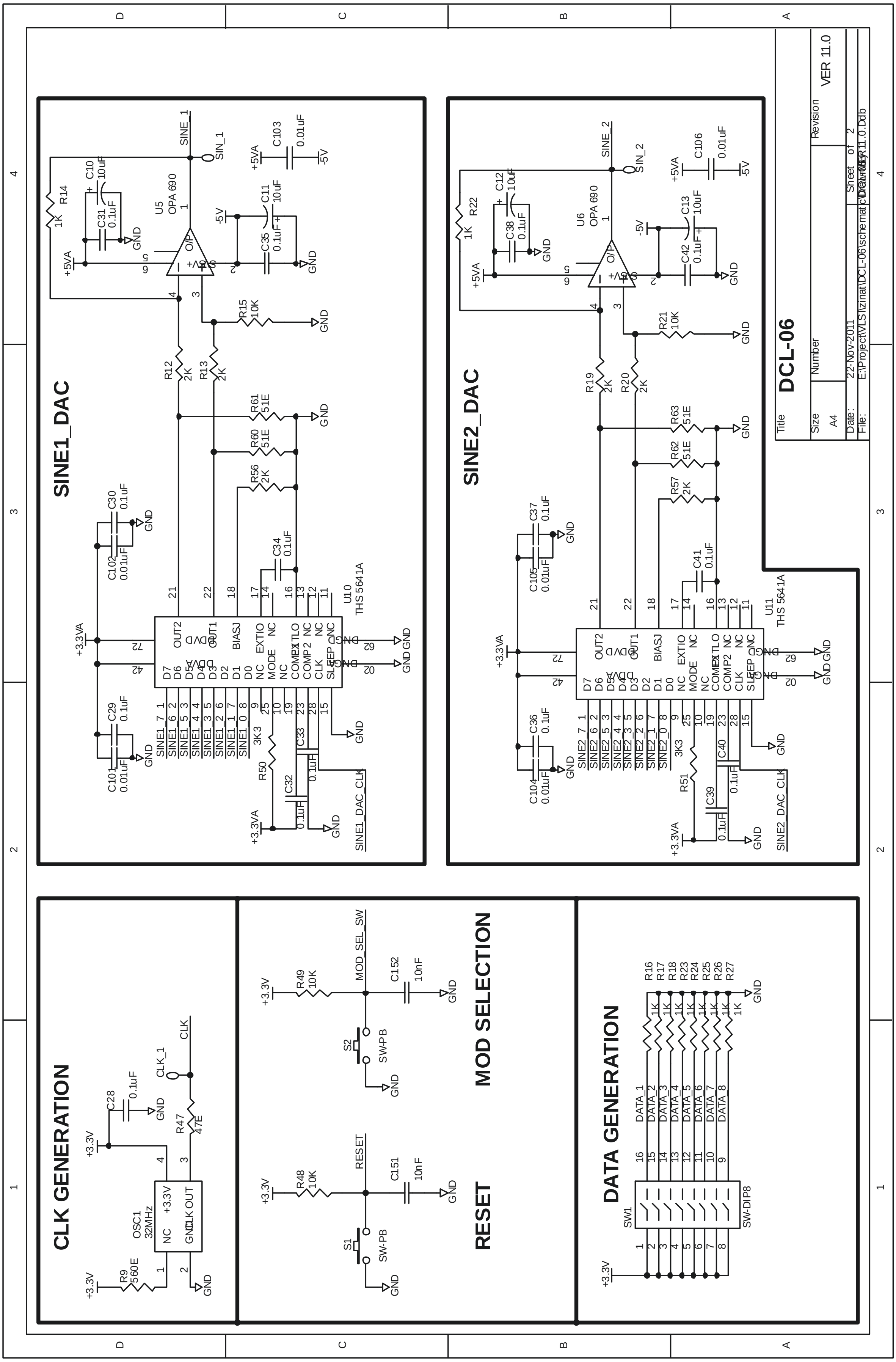
FSK modulated input is fed to the FSK demodulator. Logic for FSK demodulation is written in VHDL. FSK demodulator consists of ZERO PULSE DETECTOR AND COUNTER, THRESHOLD DETECTOR and DECISION DEVICE. ZERO PULSE DETECTOR detects the each zero crossing of FSK IN signal by giving pulses. ZERO PULSE COUNTER resets for every zero crossing for every zero crossing and would have two discrete values at the output due to two distinct frequency of FSK IN. THRESHOLD DETECTOR detects a threshold somewhere in between these two distinct values. Output of DECISION DEVICE is the recovered data.

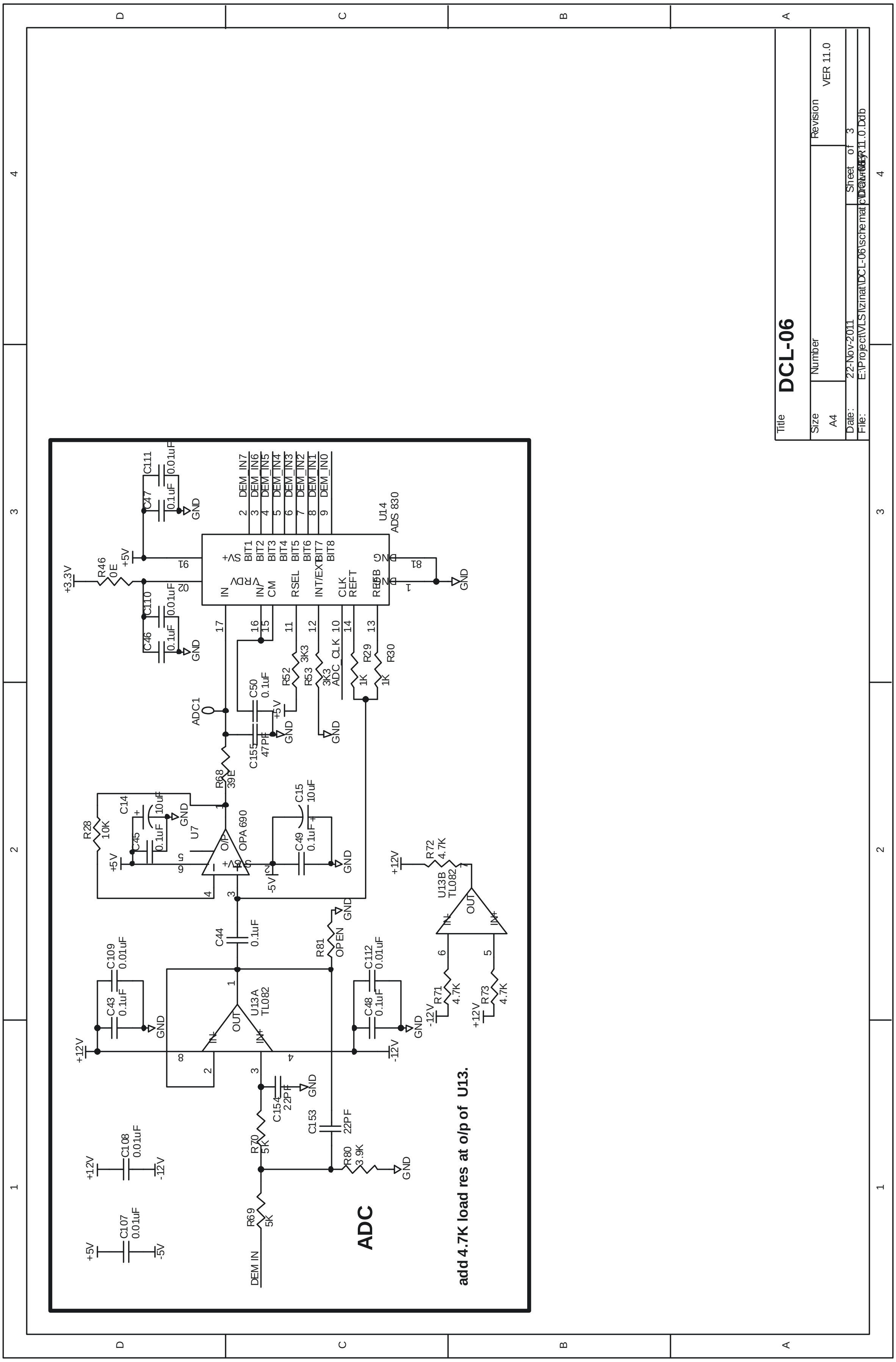
- **PSK Demodulator Section**

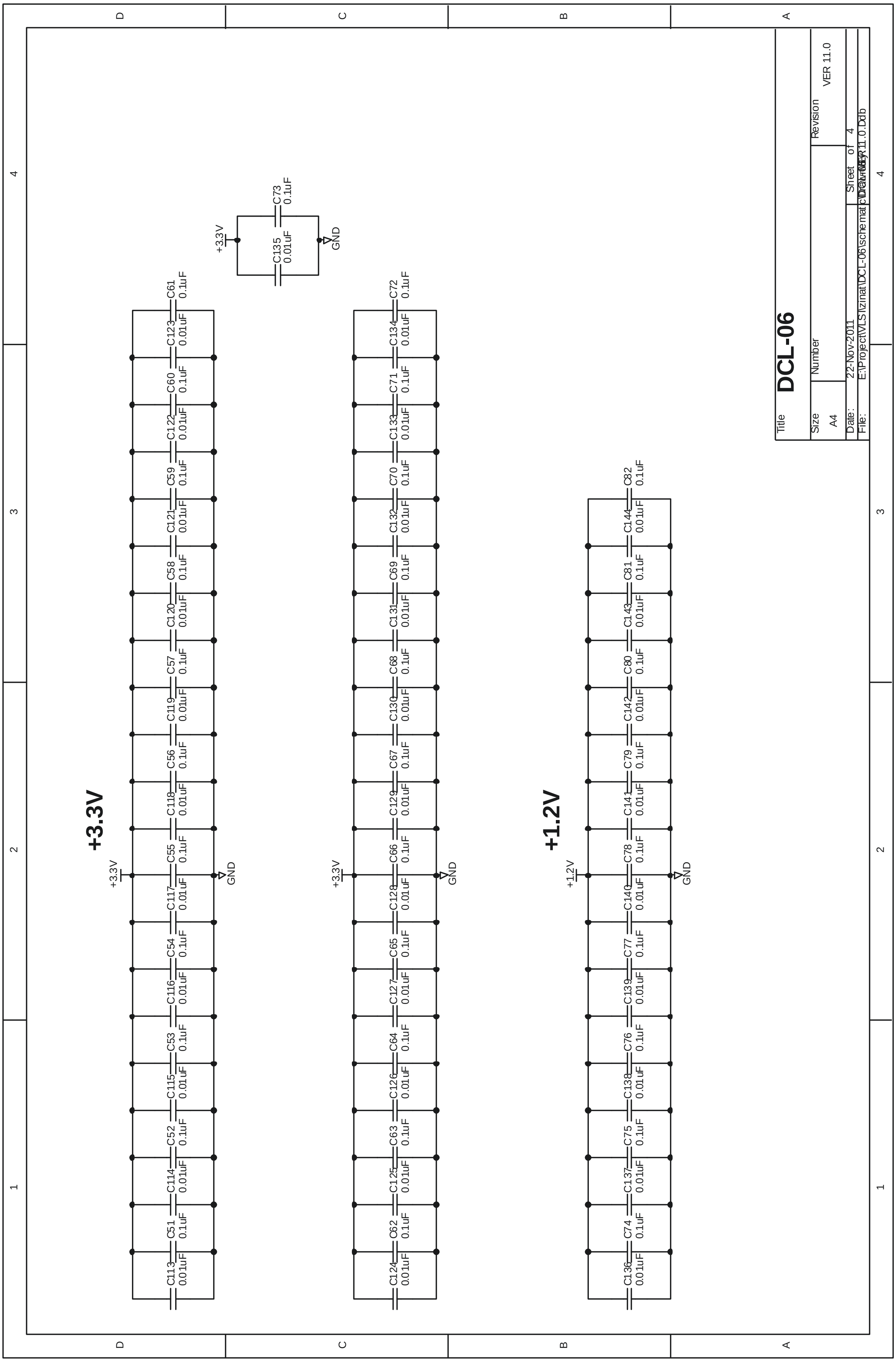
Refer sheet 5 of DCL-06 for the Circuit diagram of PSK modulator. MID_OUT_DAC section consist of 14 bit DAC THS 5671A (U15) and OPA 690 (U9). The PSK modulated input is fed to the MULTIPLIER. FSK demodulation employs squaring and filtering logic which is written in VHDL. MID_OUT_DAC section gives square of input signal due to which phase of the PSK modulated signal is removed. Output of MULTIPLIER is fed to the LOW PASS FILTER which filtered out the high frequency component and pass low frequency i.e. 256 KHz. Output of LOW PASS FILTER is the recovered data.

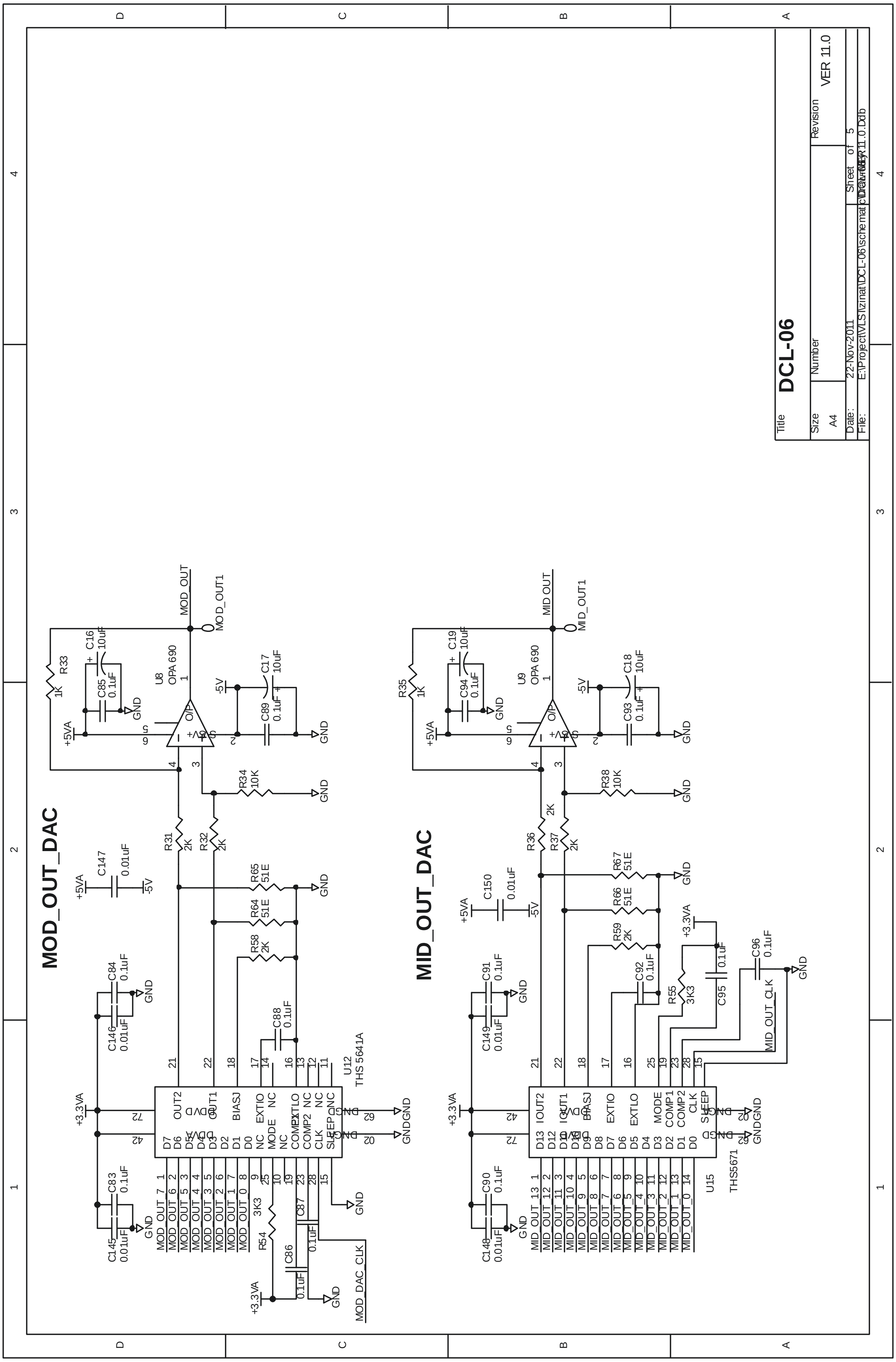
CIRCUIT DIAGRAM OF DCL-06

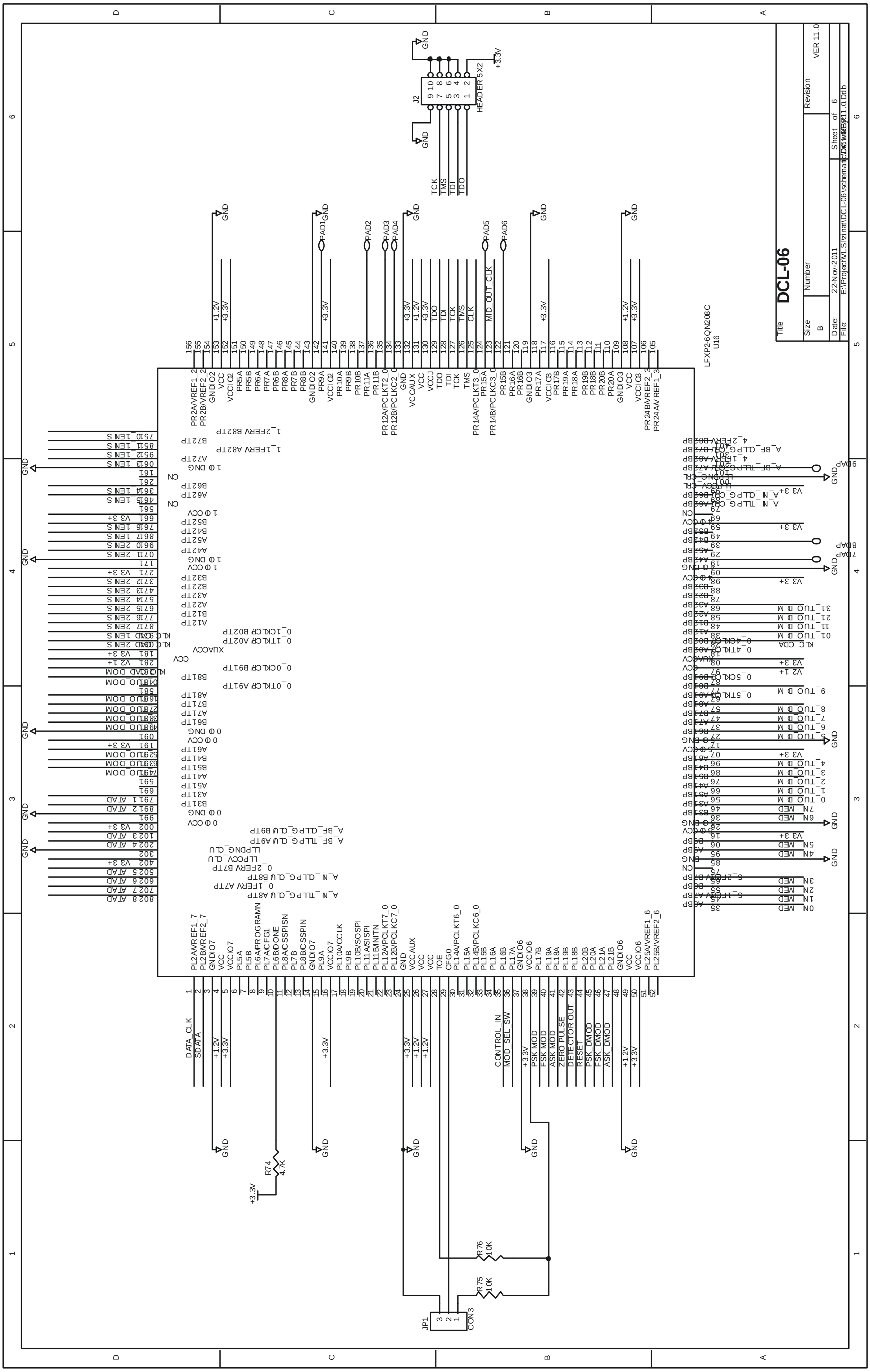


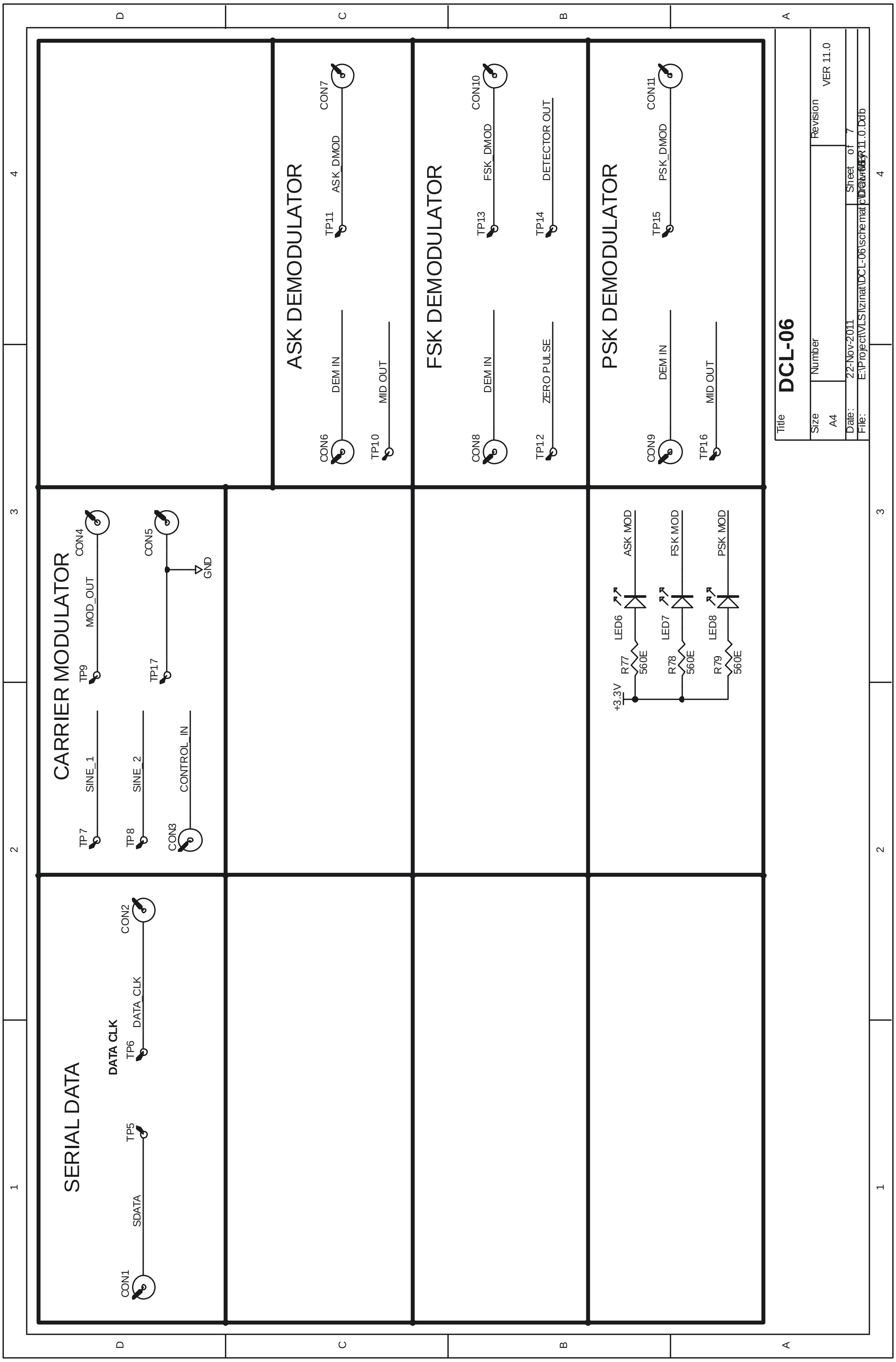

SINE1_DAC
SINE2_DAC
CLK GENERATION
RESET
MOD SELECTION
DATA GENERATION
U5 OPA 690
U6 OPA 690
U10 THS 5641A
U11 THS 5641A
OSC1 32MHz
SW1 SW-DIP8
S1 SW-PB
S2 SW-PB
Title DCL-06
Size A4
Revision VER 11.0
Date: 22-Nov-2011


ADC
add 4.7K load res at o/p of U13.
DCL-06
Title
Size
A4
Number
Revision
VER 11.0
Date: 22-Nov-2011
Sheet of 3
U14
ADS 830
U7
OPA 690
U13A
TL082
U13B
TL082
DEM IN
ADC1


+3.3V
+1.2V
GND
C73 0.1uF
C135 0.01uF
Title DCL-06
Size A4
Number
Revision VER 11.0
Date: 22-Nov-2011
Sheet of 4
File:


MOD_OUT_DAC
MID_OUT_DAC
U8
OPA 690
U9
OPA 690
U12
THS 5641A
U15
THS5671
MOD_OUT
MOD_OUT1
MID OUT
MID_OUT1
MOD_DAC_CLK
MID_OUT_CLK
Title
DCL-06
Size
A4
Number
Revision
VER 11.0
Date: 22-Nov-2011
Sheet of 5


DCL-06
VER 11.0
22-Nov-2011
LFXP2-6QN208C
U16
HEADER 5X2
J2
JP1
CON3


SERIAL DATA
CON1
SDATA
TP5
TP6
DATA CLK
DATA_CLK
CON2
CARRIER MODULATOR
TP7
SINE_1
TP8
SINE_2
CON3
CONTROL_IN
TP9
MOD_OUT
CON4
TP17
CON5
GND
ASK DEMODULATOR
CON6
DEM IN
TP10
MID OUT
TP11
ASK_DMOD
CON7
FSK DEMODULATOR
CON8
DEM IN
TP12
ZERO PULSE
TP13
FSK_DMOD
CON10
TP14
DETECTOR OUT
PSK DEMODULATOR
CON9
DEM IN
TP16
MID OUT
TP15
PSK_DMOD
CON11
+3.3V
R77
560E
LED6
ASK MOD
R78
560E
LED7
FSK MOD
R79
560E
LED8
PSK MOD
Title
DCL-06
Size
A4
Number
Revision
VER 11.0
Date:
22-Nov-2011
Sheet of 7
File:
Drawn By:

EXPERIMENT

NO. 1

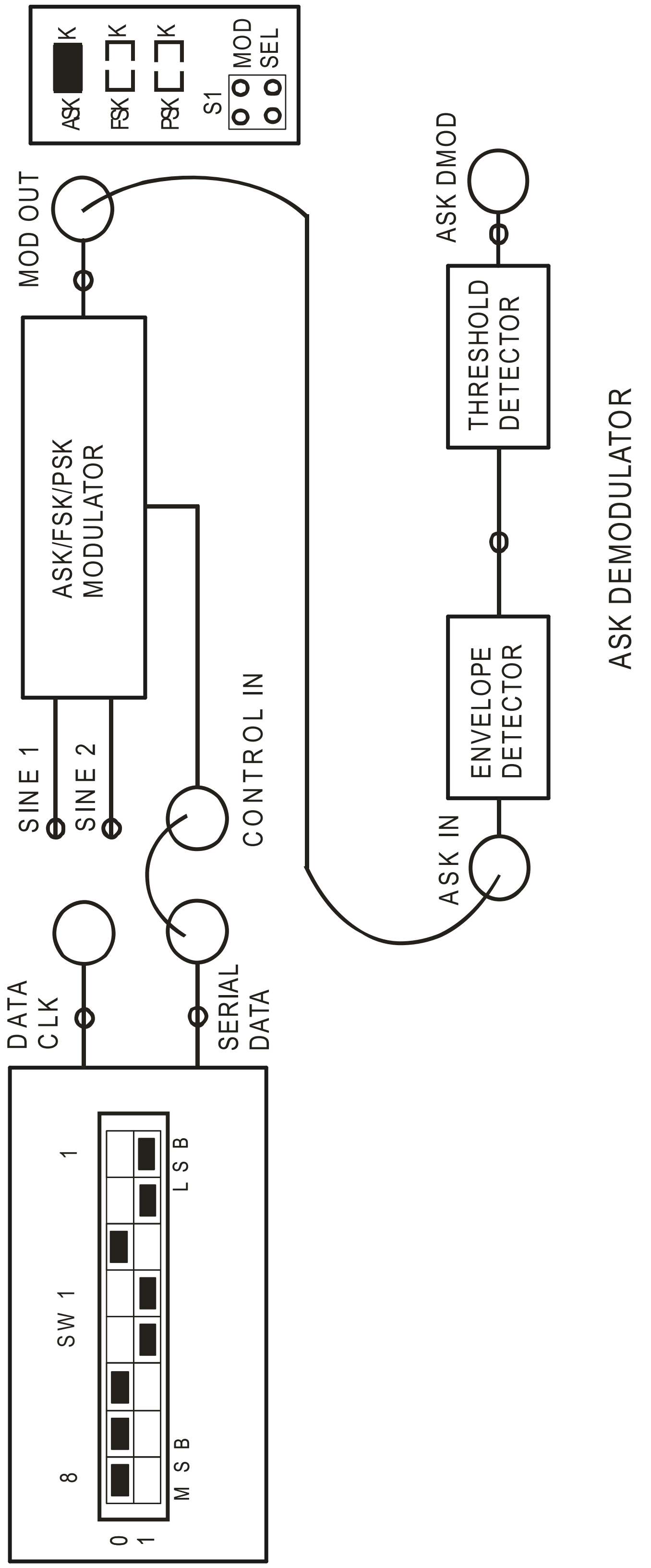


FIG 1.1 BLOCK DIAGRAM FOR ASK MODULATION AND DEMODULATION TECHNIQUES

EXPERIMENT No: 1

NAME

AMPLITUDE SHIFT KEYING MODULATION AND DEMODULATION TECHNIQUES

OBJECTIVE

Study of Carrier Modulation Techniques by Amplitude Shift Keying method

THEORY

Carrier modulation is a technique by which digital data is made to modulate a continuous wave (sine wave) carrier. For all types of carrier modulation, the carrier frequency should be atleast 2 times that of modulating frequency. In Amplitude shift keying, the carrier is transmitted when the modulating data is `one' and the carrier is rejected from transmission when the data is `zero'. In DCL-06 the ASK Modulators employs an Digital Multiplexer as a modulating switch, which can switch between carrier and ground, for every `one' to `zero' transitions. The carrier frequency chosen for ASK modulation is 1.024 MHz

ASK DEMODULATOR block on board employs an envelope detector to recover the data from the modulated carrier. The ASK modulated input is fed to the full wave rectifier. The rectified input is fed to the filter, where the original data is recovered. The threshold detector is used to recover the original amplitude levels of the data. So whenever the sine wave is transmitted, the detector identifies it as a `one' and whenever the carrier is absent, the detector identifies it as a `zero'.

EQUIPMENTS

Experimental Kit DCL-06.
Patch Chords.
Power supply.
30MHz Dual Trace Oscilloscope/DSO.

PROCEDURE

1. Refer to the Fig. 1.1 and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-06** and switch it on.
3. Connect **SERIAL DATA** generated on board to **CONTROL IN** of **CARRIER MODULATOR.**
4. Select ASK modulation using switch S1, the ASK LED will glow.
5. Observe the waveforms at SINE1, SINE2 and MOD OUT.

6. Connect ASK modulated signal **MOD OUT** to the **ASK IN** of the **ASK DEMODULATOR.**
7. Observe various waveforms as mentioned below.

OBSERVATION

Observe the following signal on the oscilloscope and plot it on the paper.

1. **SERIAL DATA** with respect to **DATA CLK**.	**FIG. 1.2(a)**
2. **SINE 1**(1.024 MHz) and **SINE 2** (no signal i.e. grounded).	**FIG. 1.2(b)**
3. ASK modulated signal **MOD OUT** with respect to **CONTROL IN**.	**FIG. 1.2(c)**
4. Output of **ENVELOPE DETECTOR** with respect to **ASK IN**.	**FIG. 1.2(d)**
5. **ASK DMOD** with respect to **CONTROL IN**.	**FIG. 1.2(e)**

CAPTURED WAVEFORM

CH 1: DATA CLK (256 KHz) & CH 2: SERIAL DATA (00011011)

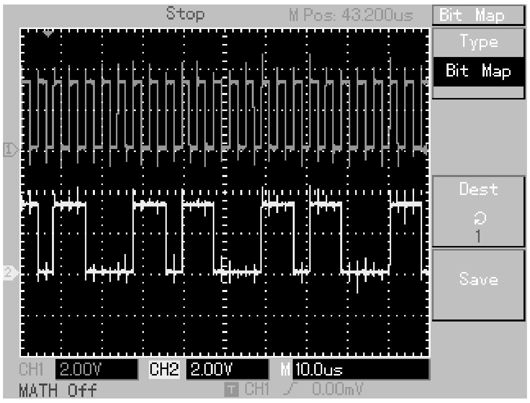


FIG. 1.2(a)

CH 1: SINE 1 (1.024MHz, 0°) & CH 2: SINE 2 (GND)

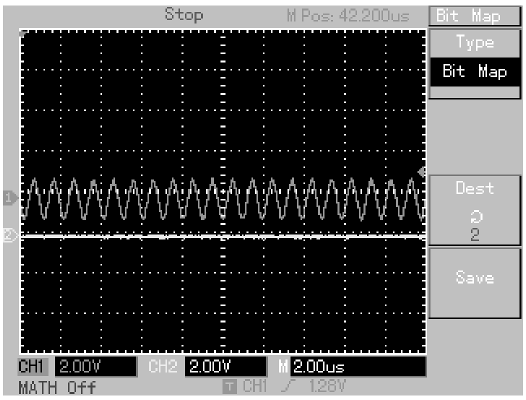


FIG. 1.2(b)

CH 1: CONTROL IN & CH 2: MOD OUT (ASK)

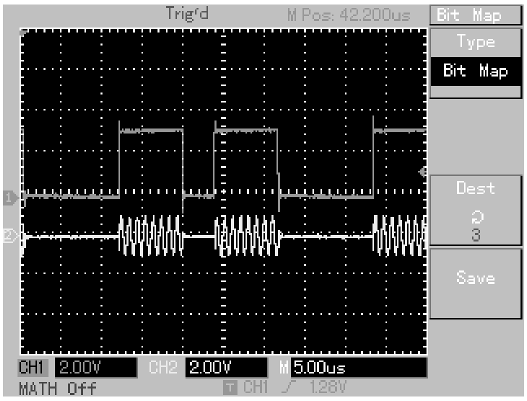


FIG. 1.2(c)

CH 1: ASK IN & CH 2: Output of ENVELOPE DETECTOR

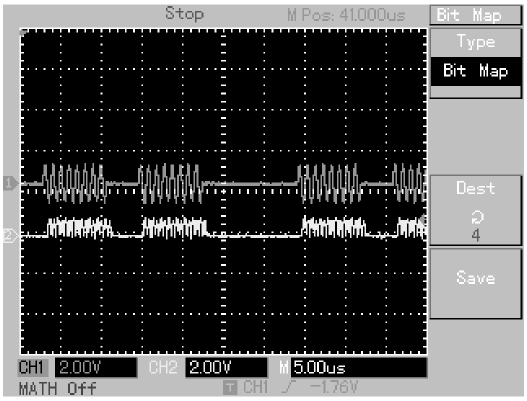


FIG. 1.2(d)

CH 1: CONTROL IN & CH 2: ASK DMOD

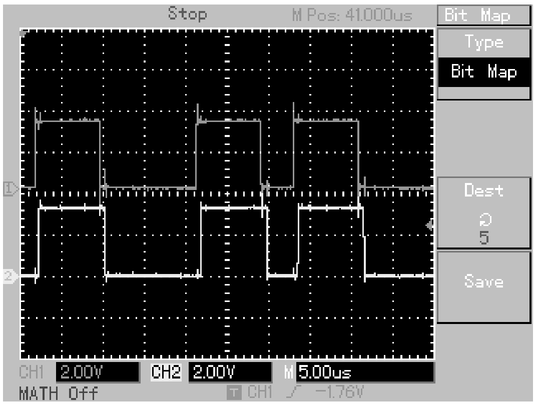


FIG. 1.2(e)

CONCLUSION:

It has been observed that a very small time lag between the modulating data and the recovered data.

EXPERIMENT

NO. 2

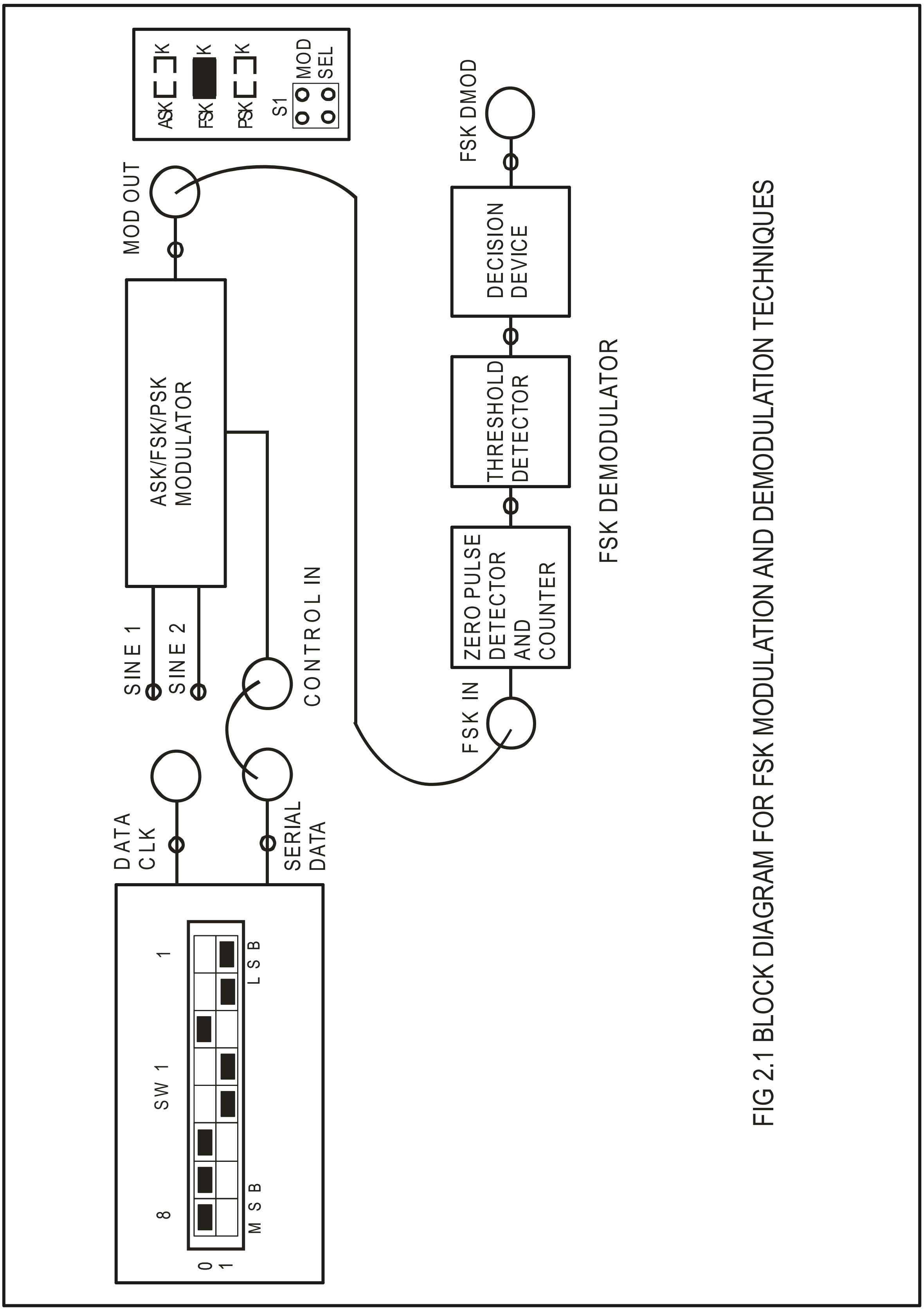


FIG 2.1 BLOCK DIAGRAM FOR FSK MODULATION AND DEMODULATION TECHNIQUES

EXPERIMENT No: 2

NAME

FREQUENCY SHIFT KEYING MODULATION AND DEMODULATION TECHNIQUES

OBJECTIVE

Study of Carrier Modulation Techniques by Frequency Shift Keying method.

THEORY

In Frequency Shift Keying modulation techniques, the modulated output shifts between two frequencies for all `one' (mark) to `zero' (space) transitions. The carrier frequency chosen for FSK modulation are 1.024 MHz and 512 KHz. Note that the above frequencies are greater than twice the modulating frequency. Note that the FSK may be thought of as an FM system in which the carrier frequency is midway between the mark and space frequencies, and modulation is by a square wave.

CARRIER GENERATOR block on board generates the carrier waves 1.024 MHz and 512 KHz, which are available at SINE 1 and SINE 2 post.

The FSK modulator is also built around the 2 to 1 Digital Multiplexer which switches between the1.024 MHz and 512 KHz signals for all 'one' to `zero'' transitions.

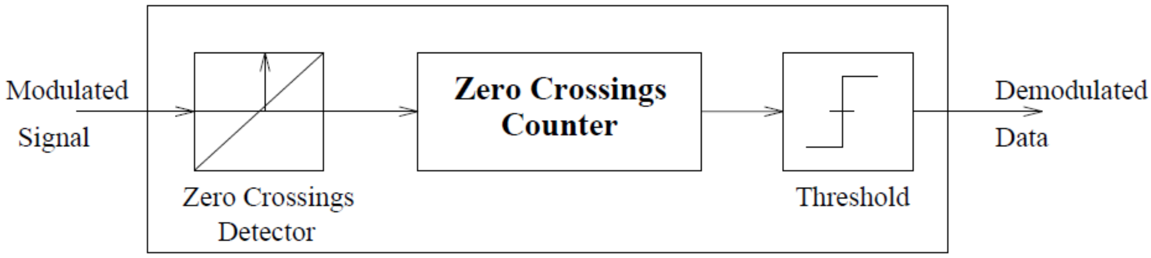


Figure 2.2: Block diagram for the FSK demodulator

FSK signals have multiple tones in them. Each tone can be characterized by the number of clock cycles it takes between two zero crossings for a given sampling frequency. For example a 1MHz signal sampled at 32 MHz will have 16 samples between two zero crossings and a 512 kHz signal would have about 32 samples. So a counter that can resets for every zero crossing would have two discreet values at the output. A threshold somewhere in between 32 and 16 would hard limit the output to digital levels. Thus demodulating the FSK signal. The block diagram for the FSK demodulator is shown in the figure to demonstrate this the two carriers are chosen to be 1MHz and 512 kHz for Mark and Space frequencies.

EQUIPMENTS

Experimental Kit DCL-06
Patch Chords
Power supply
30MHz Dual Trace Oscilloscope/DSO.

PROCEDURE

1. Refer to the Fig. 2.1 and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-06** and switch it on.
3. Connect **SERIAL DATA** generated on board to **CONTROL INPUT** of **CARRIER MODULATOR.**
4. Select FSK modulation using switch S1, the FSK LED will glow.
5. Observe the waveforms at SINE 1,SINE 2 and MOD OUT.
6. Connect FSK modulated signal **MOD OUT** to the **FSK IN** of the **FSK DEMODULATOR**.
7. Observe various waveforms as mentioned below.

OBSERVATION

Observe the following signal on the oscilloscope and plot it on the paper.

1. **SERIAL DATA** with respect to **DATA CLK**. **FIG. 2.3(a)**
2. **SINE 1**(1.024 MHz) **and SINE 2** (512 KHz). **FIG. 2.3(b)**
3. **FSK** modulated signal **MOD OUT** with respect to **CONTROL IN**. **FIG. 2.3(c)**
4. Output of Zero Pulse Detector at its test point with respect to **FSK IN**. **FIG. 2.3(d)**
5. Output of Threshold Detector at its test point with respect to **FSK IN**. **FIG. 2.3(e)**
6. **FSK DMOD** with respect to **CONTROL IN**. **FIG. 2.3(f)**

CAPTURED WAVEFORM

CH 1: DATA CLK (256 KHz) & CH 2: SERIAL DATA (00011011)

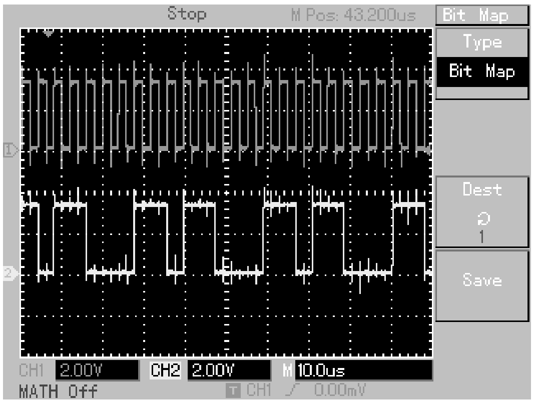


FIG. 2.3(a)

CH 1: SINE 1 (1.024MHz, 0°) & CH 2: SINE 2 (512 KHz, 0°)

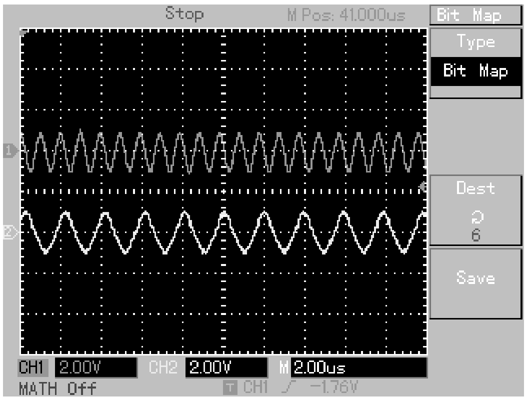


FIG. 2.3(b)

CH 1: CONTROL IN & CH 2: MOD OUT (FSK)

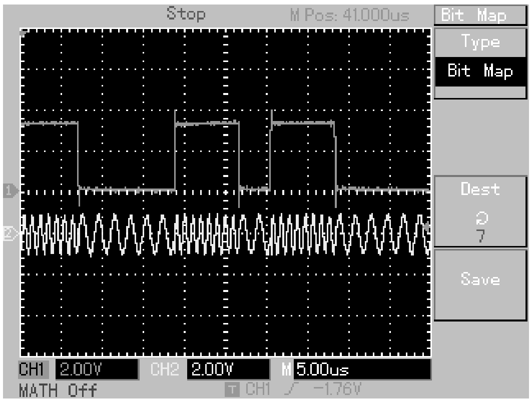


FIG. 2.3(c)

CH 1: FSK IN & CH 2: ZERO PULSE DETECTOR OUT

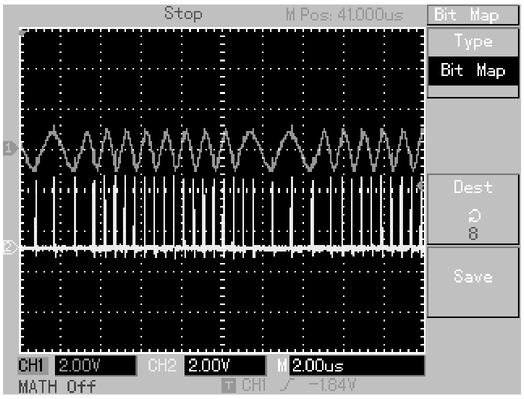


FIG. 2.3(d)

CH 1: ZERO PULSE DETECTOR OUT & CH 2: THRESHOLD DETECTOR OUT

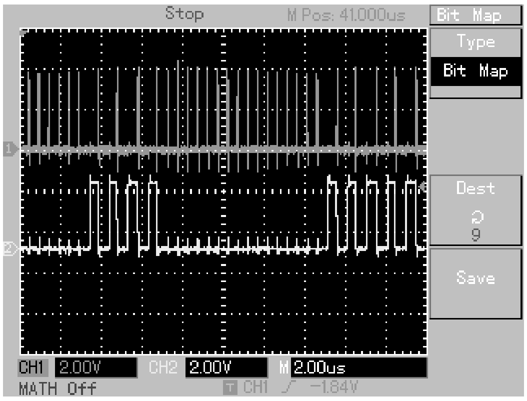


FIG. 2.3(e)

CH 1: CONTROL IN & CH 2: FSK DMOD

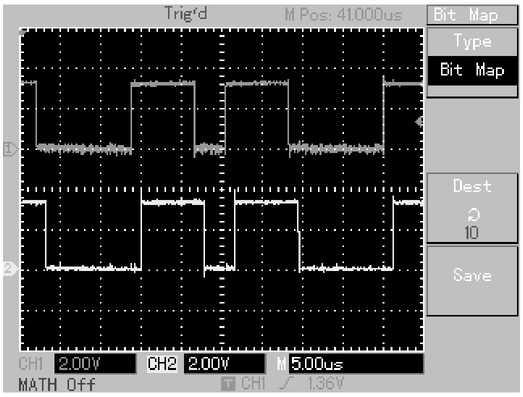


FIG. 2.3(f)

<u>CONCLUSION</u>:

A small phase lag exists between the modulating data and the recovered data because of internal delay of FPGA.

EXPERIMENT

NO. 3

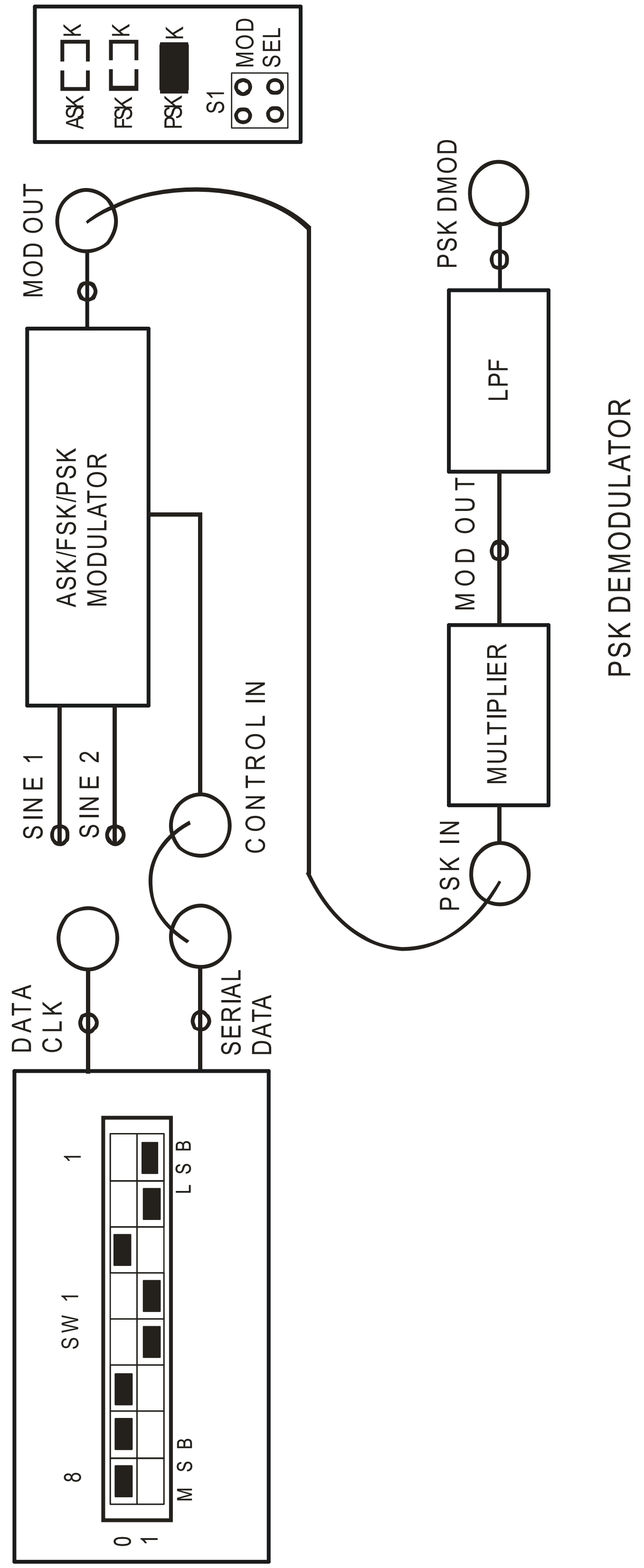


FIG 3.1 BLOCK DIAGRAM FOR PSK MODULATION AND DEMODULATION TECHNIQUES

EXPERIMENT No: 3

NAME

PHASE SHIFT KEYING MODULATION AND DEMODULATION TECHNIQUES.

OBJECTIVE

Study of Carrier Modulation Techniques by Phase Shift Keying method.

THEORY

In Phase Shift Keying (PSK) modulation techniques, the modulated output switches between in-phase and out-of phase component of the carrier frequency for all `one' to `zero' transitions. The carrier frequency chosen for PSK modulation are 1.024MHz (0°) and 1.024MHz (180°).

CARRIER GENERATOR block on board generates the carrier waves 1.024MHz (0°) and 1.024MHz (180°), which are available at SINE 1 and SINE 2 post. The PSK modulator is also built around the 2 to 1 Multiplexer which switches between the 1.024MHz (0°) and 1.024MHz (180°)signals for all 'one' to `zero'' transitions occurring in the transmitted data stream.

For PSK demodulation the incoming signal is applied to the multiplier to remove the phase shift. The output of the multiplier is then given to Low Pass Filter, where we get the recovered transmitted data. These recovered data having exactly same phase & frequency compared to transmitted data

EQUIPMENTS

Experimental Kit DCL-06.
Patch Chords.
Power supply.
30MHz Dual Trace Oscilloscope/DSO.

PROCEDURE

1. Refer to the Fig. 3.1 and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-06** and switch it on.
3. Connect **SERIAL DATA** generated on board to **CONTROL INPUT** of **CARRIER MODULATOR.**
4. Select PSK modulation using switch S1, the **PSK LED** will glow.
5. Observe the waveforms at **SINE 1, SINE 2** and **MOD OUT**.

6. Connect PSK modulated signal **MOD OUT** to the **PSK IN** of the **PSK DEMODULATOR.**
7. Observe various waveforms as mentioned below.

OBSERVATION

Observe the following signal on the oscilloscope and plot it on the paper.

1. **SERIAL DATA** with respect to **DATA CLK**. **FIG. 3.2(a)**
2. **SINE 1**(1.024 MHz, 0°) and **SINE 2** (1.024 MHz, 180°). **FIG. 3.2(b)**
3. **PSK** modulated signal **MOD OUT** with respect to **CONTROL IN**. **FIG. 3.2(c)**
4. **MID OUT** with respect to **PSK IN**. **FIG. 3.2(d)**
5. **PSK DMOD** with respect to **CONTROL IN**. **FIG. 3.2(e)**

CAPTURED WAVEFORM

CH 1: DATA CLK (256 KHz) & CH 2: SERIAL DATA (00011011)

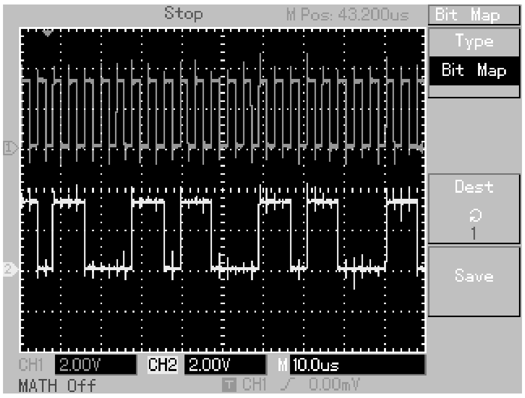


FIG. 3.2(a)

CH 1: SINE 1 (1.024MHz, 0°) & CH 2: SINE 2 (1.024MHz, 180°)

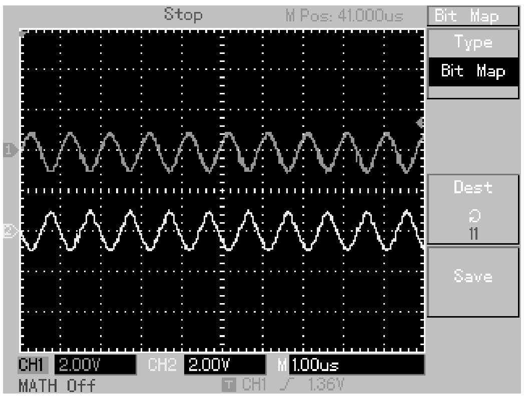


FIG. 3.2(b)

CH 1: CONTROL IN & CH 2: MOD OUT (PSK)

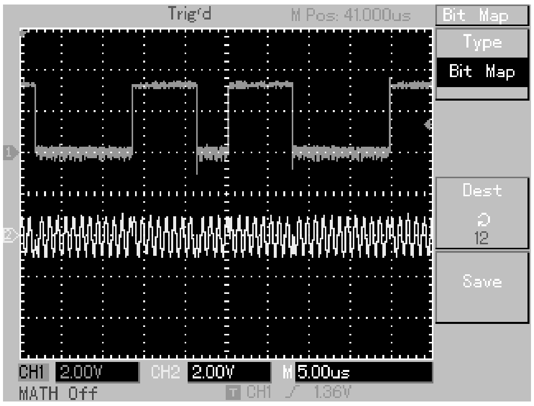


FIG. 3.2(c)

CH 1: CONTROL IN & CH 2: MID OUT

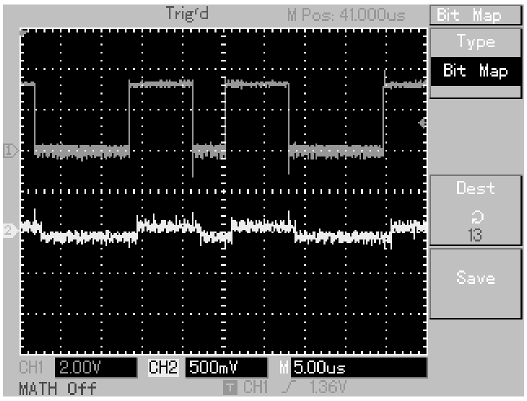


FIG. 3.2(d)

CH 1: CONTROL IN & CH 2: PSK DMOD

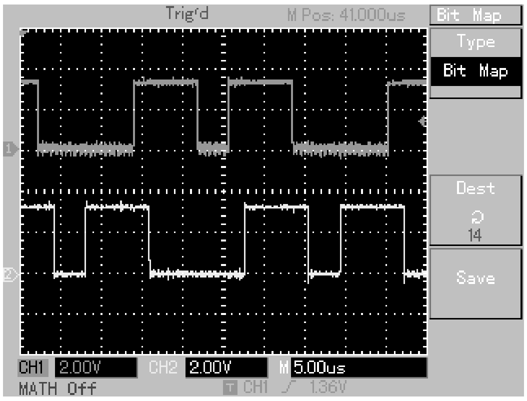


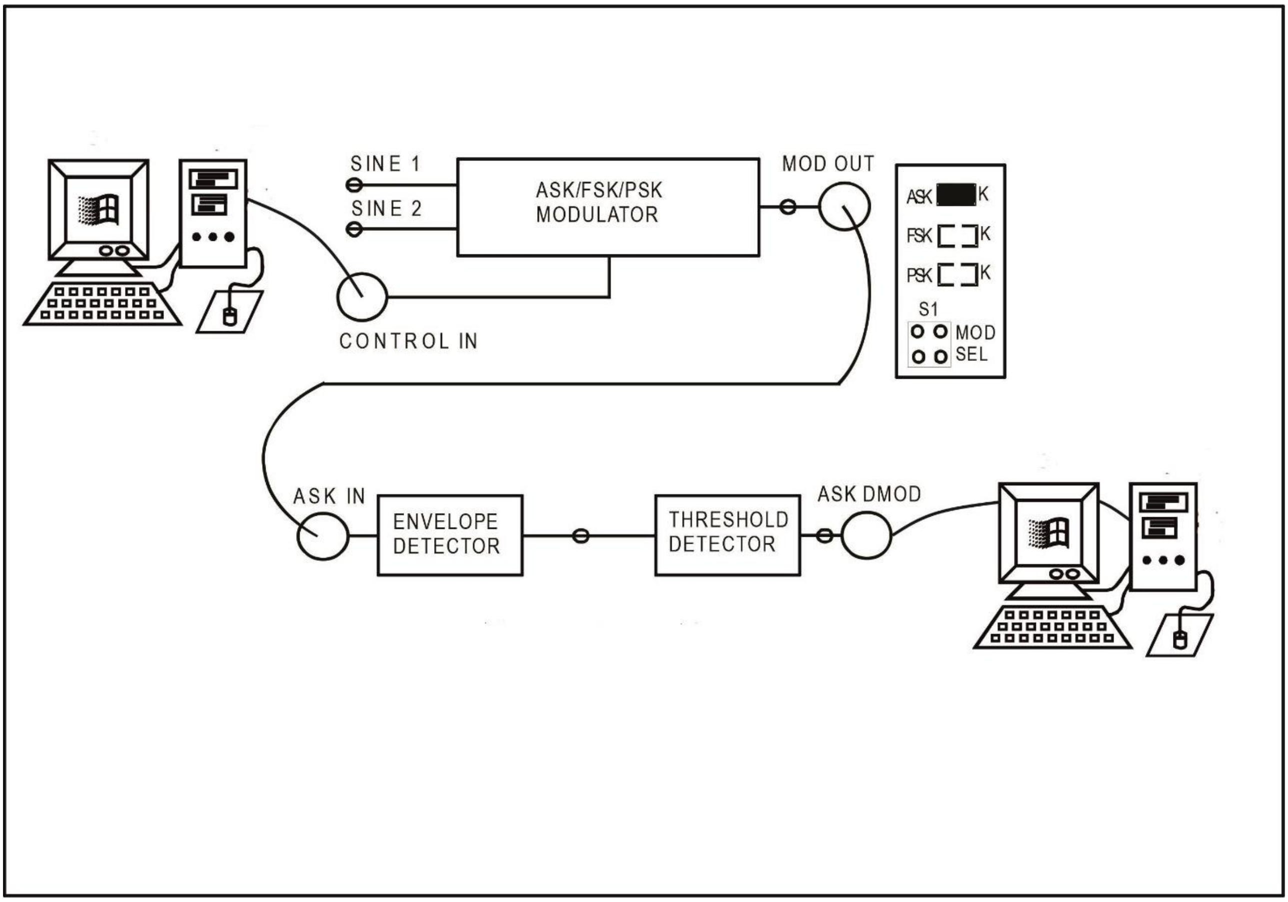
FIG. 3.2(e)

CONCLUSION:

It is observed that the successful operation of the PSK Demodulator is fully dependent on the phase components of the transmitted modulated carrier. If the phase reversal of the modulated carrier along with the rising and falling edges of the data is not proper, then the efficient detection of data from PSK modulated carrier becomes impossible.

EXPERIMENT

NO. 4


SINE 1
SINE 2
ASK/FSK/PSK
MODULATOR
MOD OUT
CONTROL IN
ASK
FSK
PSK
S1
MOD
SEL
ASK IN
ENVELOPE
DETECTOR
THRESHOLD
DETECTOR
ASK DMOD

EXPERIMENT No: 4

NAME

PC TO PC COMMUNICATION USING ASK /FSK /PSK MODULATION AND DEMODULATION TECHNIQUES.

OBJECTIVE

Study of PC to PC communication using ASK/FSK/PSK Modulation and De-Modulation Techniques.

THEORY

The proposes an innovative experimentation for communication between two computers. The proposed technique uses ASK /FSK / PSK modulator and demodulator. A data from one PC can be transferred to another PC in serial form by using Hyper Terminal and USB to TTL converter. The purpose of design of this experiment is to study the basic technical concepts of interfacing analog ICs and electronic circuits with PC, use of USB to TTL converter, serial data transmission of PC, use of Hyper Terminal, understanding the concept of Acoustic Coupling, by using the existing set up in the lab.

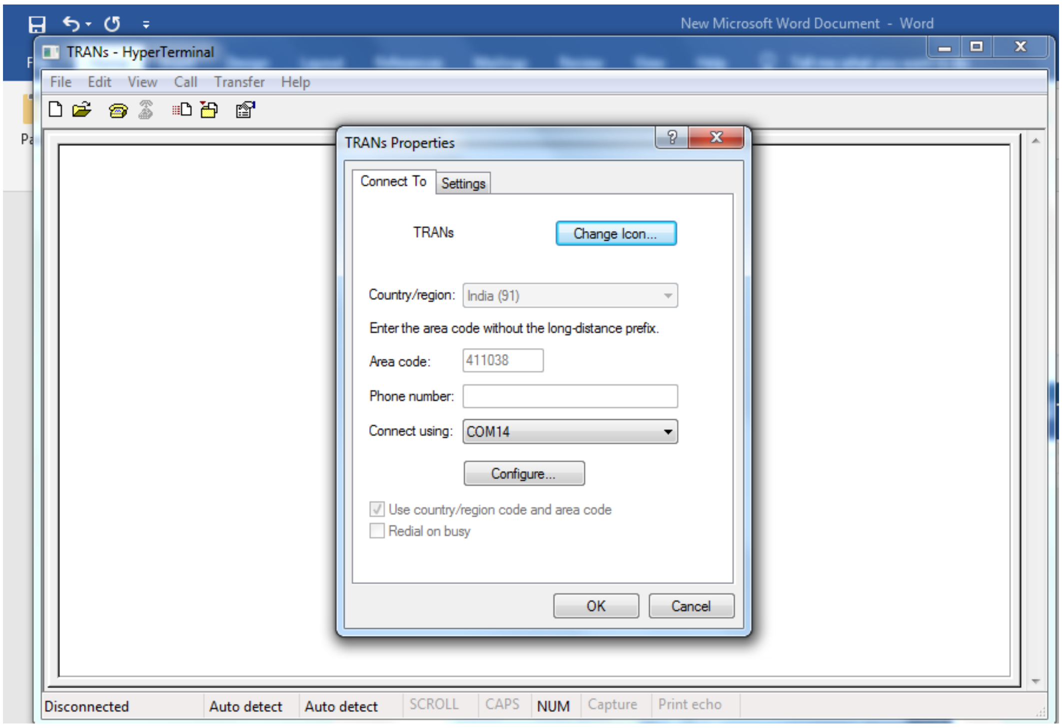
EQUIPMENTS

- DCL-06 Kit.
- Patch Chords.
- Power supply.
- USB Cable_02 nos.
- Connecting links

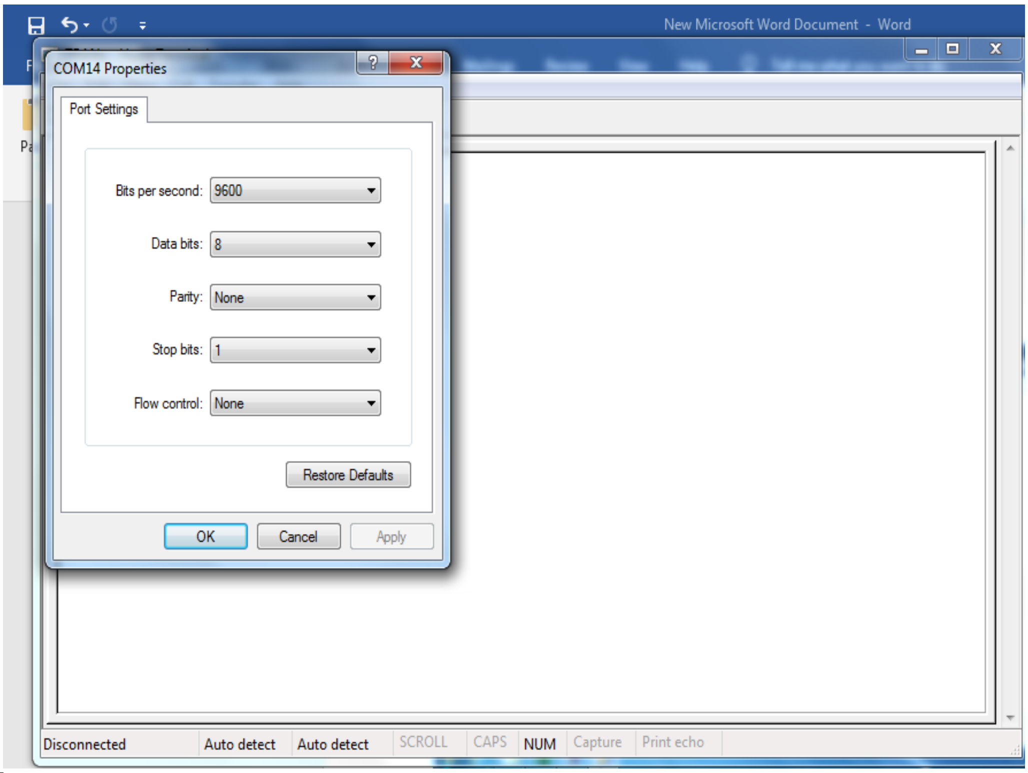
PROCEDURE

1. Refer to the Fig. 4.1 and carry out the following connections and switch setting.
2. Connect power supply in proper polarity to the kit **DCL-06** and switch it on.
3. Connect One PC with connecting link and USB cable to **CONTROL INPUT** of **CARRIER MODULATOR.**
4. Select ASK or FSK or PSK modulation using switch S1.
5. Connect ASK or FSK or PSK modulated signal **MOD OUT** to the respective **DEMODULATOR INPUT.**
6. Connect **DEMOD OUTPUT** to the second PC with USB cable.
7. Now check USB port in Device Manager.
8. Open HYPER TERMININAL.
9. Set all the parameter as shown in below.

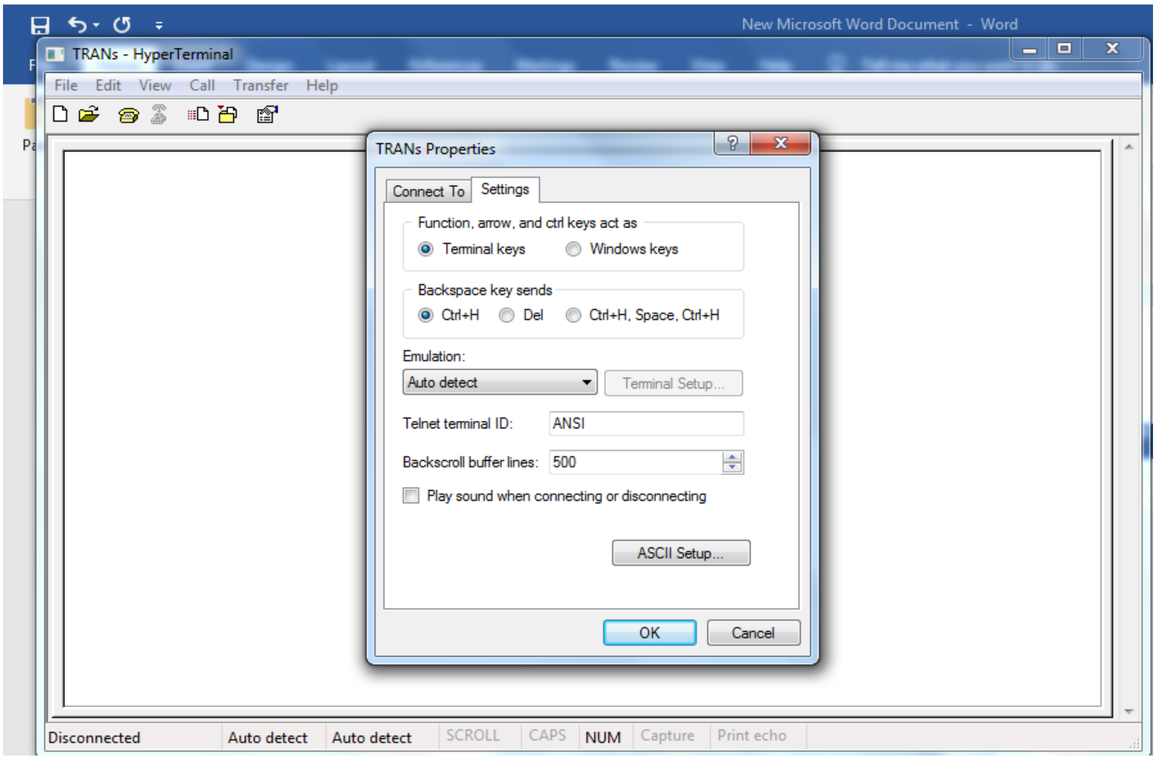
- Set The COM PORT for first PC



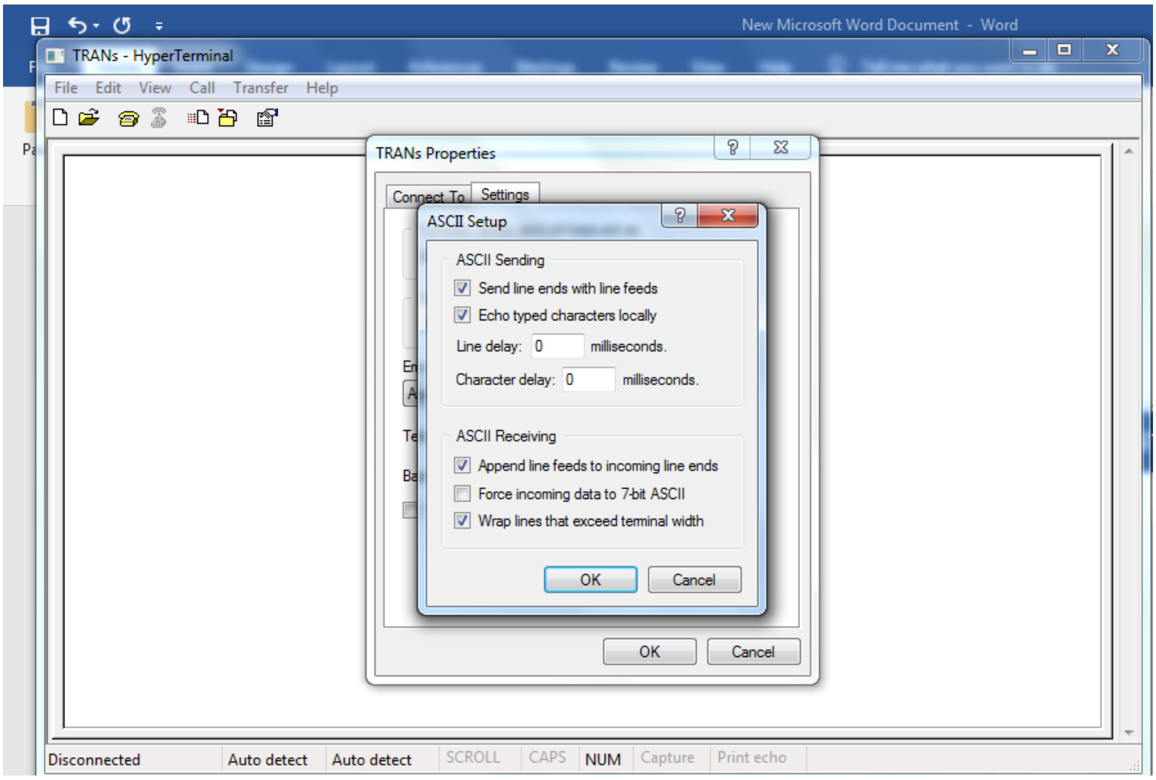
- Set Bits per second_9600 , Data Bits_8, Parity None ,Stop bit_1, Flow Control None & then press **OK**



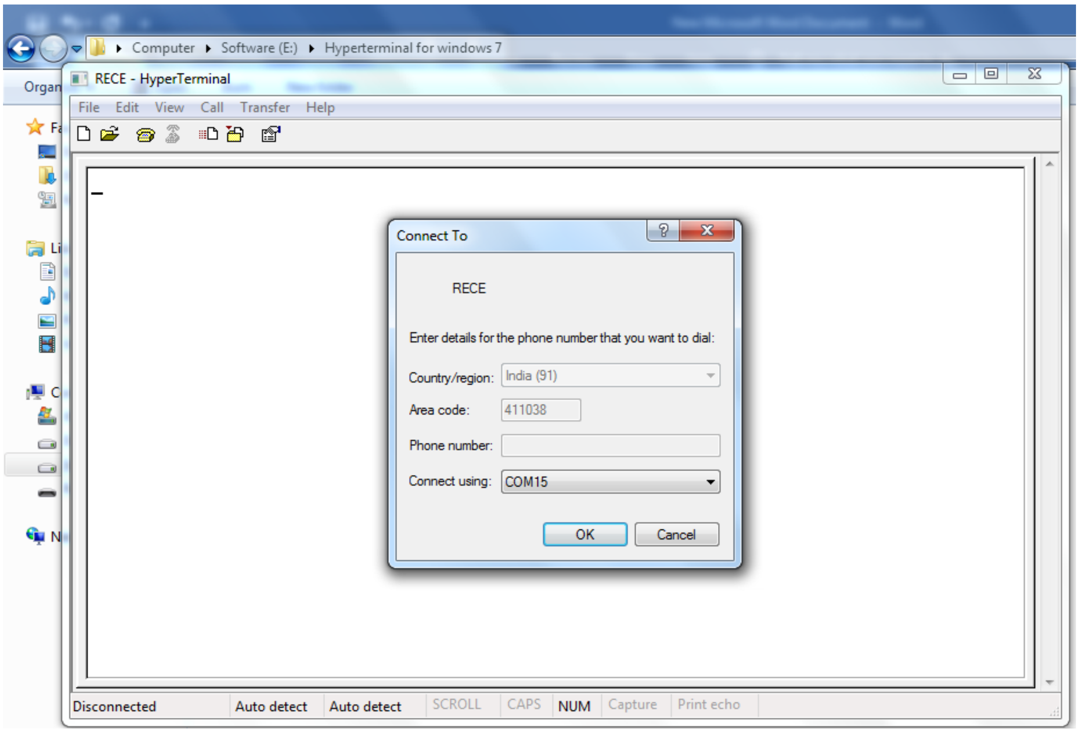
- Click on **Setting** and click on **ASCII Setup.**



- Tick mark on **Send line ends with line feeds**, **Echo type characters locally** and **Append line feeds to incoming line ends.**



- Set the same setting for second PC



OBSERVATION

Now change baud rate and modulation technique.